KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 9 STANDARD

CHAPTER 10

YEAR 9 MATHEMATICS

Surface area and volume

10 lessons · two pages each · Easy, Medium, then Mixed practice

10.1 Metric units
10.2 Precision and absolute error
10.3 Percentage error
10.4 Perimeter and area of composite shapes
10.5 Areas of quadrilaterals
10.6 Circumference and arc length
10.7 Areas of circles and sectors
10.8 Surface area of a prism
10.9 Surface area of a cylinder
10.10 Volumes of prisms and cylinders

NAME _____ CLASS _____ TEACHER _____

Artwork: a silver spiral on the western edge of Seahorse Valley, coloured in purple and silver, part of the Mandelbrot set.

How to use this booklet

1. Each lesson is **two pages**. Read the **learning intention** and tick each **success criterion** when you can do it.

2. Page 1 is **Easy** and page 2 is **Medium**. Each page starts with an example, then questions for you.

WE DO **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Your turn. Work down the box one round at a time. Answers go on the line; show working in the boxes.

3. **Mixed practice** ends page 2: today's skill mixed with earlier lessons. The tag on each question (like **10.2** or **Year 8**) says where it comes from.

4. Stuck? Read the **Remember** line in the blue box, then look back at the example or the lesson in the tag. Answers with π: **one decimal place** unless the question says otherwise.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
10.1	Metric units	3		☐
10.2	Precision and absolute error	5		☐
10.3	Percentage error	7		☐
10.4	Perimeter and area of composite shapes	9		☐
10.5	Areas of quadrilaterals	11		☐
10.6	Circumference and arc length	13		☐
10.7	Areas of circles and sectors	15		☐
10.8	Surface area of a prism	17		☐
10.9	Surface area of a cylinder	19		☐
10.10	Volumes of prisms and cylinders	21		☐

In this chapter you will

- convert between metric units, including very large and very small units
- find the precision, absolute error and percentage error of a measurement
- find the perimeter and area of composite shapes and quadrilaterals
- find the circumference, arc length and area of circles, sectors and composite shapes
- find the surface area of prisms and cylinders
- find the volume and capacity of prisms and cylinders



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a silver spiral on the western edge of Seahorse Valley, coloured in purple and silver. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON 10.1

Metric units

Date _____

Learning intention
We are learning to convert between metric units, and between units of time and digital storage.

Success criteria
- ☐ I can convert units of length, mass and capacity
- ☐ I can convert units of time
- ☐ I can use the prefixes mega-, giga- and micro-

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 **Convert each one. To a smaller unit, multiply. To a bigger unit, divide.**

a 8 km to m

b 250 cm to m

c 3.5 kg to g

d 4 500 mL to L

YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER 1 km = 1000 m · 1 m = 100 cm · 1 cm = 10 mm · 1 t = 1000 kg · 1 kg = 1000 g · 1 g = 1000 mg · 1 kL = 1000 L · 1 L = 1000 mL

ROUND 1 Questions 1–8: convert each length.

1. 6 km to m
2. 3 000 m to km
3. 4 m to cm
4. 700 cm to m
5. 9 cm to mm
6. 120 mm to cm
7. 2.5 km to m
8. 45 cm to m

ROUND 2 Questions 9–16: convert each mass.

9. 5 kg to g
10. 2 000 g to kg
11. 3 t to kg
12. 6 500 kg to t
13. 0.8 kg to g
14. 750 g to kg
15. 4 000 mg to g
16. 1.2 t to kg

ROUND 3 Questions 17–24: convert each capacity.

17. 7 L to mL
18. 3 000 mL to L
19. 2 kL to L
20. 4 500 L to kL
21. 1.5 L to mL
22. 250 mL to L
23. 0.6 kL to L
24. 80 mL to L

ROUND 4 Questions 25–32: convert each one. Some take two steps.

25. 4 km to cm
26. 3 m to mm
27. 60 000 cm to km
28. 5 000 mm to m
29. 2 kg to mg
30. 1 kL to mL
31. 0.5 km to cm
32. 80 mm to m

Level 2 · Building up

Medium · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **Convert each one. Hours and minutes are not metric: 1 h = 60 min and 1 min = 60 s.**

a 3.5 h to min

b 150 s to min

c 4.2 GB to MB

d 2 500 000 µm to m

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

18 QUESTIONS

REMEMBER mega- (M) = 1 000 000 · giga- (G) = 1 000 000 000 · tera- (T) = 1000 giga- · micro- (µ) = one-millionth · 1 s = 1000 ms

ROUND 1 Questions 1–8: convert each time.

1. 5 min to s
2. 4 h to min
3. 180 s to min
4. 2.5 h to min
5. 90 min to h
6. 3 days to h
7. 4 000 ms to s
8. 1.5 min to s

ROUND 2 Questions 9–16: convert using the prefixes.

9. 3 GB to MB
10. 4 500 MB to GB
11. 2.4 TB to GB
12. 34 000 B to kB
13. 2.6 Mm to m
14. 0.7 s to ms
15. 3 000 000 µm to m
16. 5 ms to µs

ROUND 3 Questions 17–18: solve each problem. Show your working.

17. A phone has 64 GB of free space. Each video is 800 MB. How many videos fit?

18. A movie runs for 135 minutes. Write this in hours and minutes.

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

8 QUESTIONS

1. Convert 0.75 km to m. [10.1]
2. Find the area of a rectangle 8 cm long and 5 cm wide. [Year 8]
3. Which is longer: 1.2 m or 115 cm? [10.1]
4. Round 3.476 to one decimal place. [Year 8]
5. How many seconds are in 2 hours? [10.1]
6. Find the perimeter of a square with sides of 7 m. [Year 8]
7. A 2 L jug is filled with 400 mL cups of water. How many cups does it take? [10.1]
8. Convert 2.5 GB to MB. [10.1]

LESSON 10.2

Precision and absolute error

Date _____

Learning intention
We are learning to find the precision and absolute error of a measuring instrument or a measurement.

Success criteria
- ☐ I can find the precision: the smallest unit on the scale
- ☐ I can find the absolute error: $\frac{1}{2}$ of the precision
- ☐ I can find the smallest and largest possible true value

Level 1 · Getting started

Easy · 20 questions

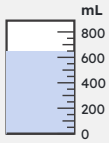
WE DO With your teacher: copy what they write on the board.

Example 1 Find the precision and the absolute error of each.

a This ruler



b This jug



c A path 24 m long, to the nearest metre

d A bag of flour with a mass of 3.6 kg

YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

20 QUESTIONS

REMEMBER The precision of a measurement is the place value of its last digit: 3.6 kg is to the nearest 0.1 kg. Absolute error = precision ÷ 2.

ROUND 1 Questions 1–8: write the precision of each measurement.

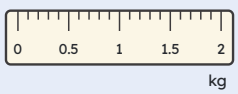
1. 7 m
2. 3.6 cm
3. 12.45 kg
4. 58 mm
5. 2.8 L
6. 0.75 t
7. 104 g
8. 9.3 s

ROUND 2 Questions 9–16: write the absolute error of each measurement.

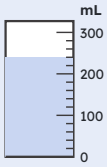
9. 15 m
10. 6.2 kg
11. 43 mL
12. 8.25 m
13. 125 cm
14. 0.4 L
15. 19.75 s
16. 312 g

ROUND 3 Questions 17–20: write the precision and the absolute error of each scale.

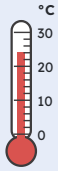
17. Kitchen scales



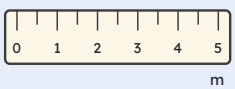
18. Jug



19. Thermometer



20. Tape measure



Level 2 · Building up

Medium · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **Limits of accuracy: measurement – absolute error to measurement + absolute error.**

a A plank is 2.45 m long. Find its precision and absolute error.

b Write the limits of the plank's length in part a.

c Jo is 162 cm tall, to the nearest cm. Write the limits.

d A jug marked every 50 mL shows 650 mL. Write the limits.

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

18 QUESTIONS

REMEMBER Smallest = measurement − absolute error · Largest = measurement + absolute error

ROUND 1 Questions 1–8: write the absolute error of each measurement.

1. 3.25 kg
2. 0.8 m
3. 14.6 L
4. 7.125 km
5. 0.04 g
6. 2.05 m
7. 120.5 cm
8. 6.375 t

ROUND 2 Questions 9–16: write the smallest and largest possible true value.

9. 24 m
10. 6.4 cm
11. 153 g
12. 5.75 kg
13. 9.8 s
14. 72 kg
15. 1.6 L
16. 0.45 m

ROUND 3 Questions 17–18: solve each problem. Show your working.

17. Kai weighs 250 g of flour on scales marked every 10 g. Write the limits of the mass.

18. A rope is 12.5 m long, to the nearest 0.1 m. Could it really be 12.56 m? Explain.

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

8 QUESTIONS

1. What is the precision of 4.75 m? 10.2
2. Convert 3.2 km to m. 10.1
3. A jug is marked every 10 mL. Absolute error? 10.2
4. Find 5% of 60. Year 8
5. Write the limits of 18 cm. 10.2
6. Convert 750 mL to L. 10.1
7. Scales are marked every 20 g. Absolute error? 10.2
8. What is the absolute error of 6.3 L? 10.2

LESSON 10.3

Percentage error

Date _____

Learning intention
We are learning to calculate the percentage error of a measurement, and use it to compare accuracy.

Success criteria
- ☐ I can find the absolute error of a measurement
- ☐ I can calculate percentage error = absolute error ÷ measurement × 100%
- ☐ I can decide which measurement is more accurate

Level 1 · Getting started

Easy · 24 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the percentage error of each measurement.

a 20 m, to the nearest metre

b 4 kg, to the nearest kilogram

c 2.5 L

d 50 cm, to the nearest centimetre

YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER Step 1: absolute error = precision ÷ 2. Step 2: percentage error = absolute error ÷ measurement × 100%.

ROUND 1 Questions 1–8: find the percentage error. Each is to the nearest whole unit.

1. 10 m
2. 25 cm
3. 50 kg
4. 4 m
5. 20 s
6. 100 g
7. 5 L
8. 40 mm

ROUND 2 Questions 9–16: find the percentage error of each measurement.

9. 0.25 m
10. 0.5 kg
11. 1.6 L
12. 12.5 cm
13. 0.2 km
14. 0.4 t
15. 6.25 s
16. 0.8 m

ROUND 3 Questions 17–24: find the percentage error, correct to one decimal place.

17. 7 m
18. 15 cm
19. 9 kg
20. 30 s
21. 3.7 L
22. 0.6 m
23. 45 g
24. 2.3 km

Level 2 · Building up

Medium · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **Use a calculator. Give each answer correct to 2 decimal places.**

a A yacht is 24 m long. Find its absolute error and percentage error.

b A parcel has a mass of 3.25 kg. Find its percentage error.

c A jug marked every 25 mL shows 375 mL. Find the percentage error.

d Which of a, b and c is the most accurate? Why?

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

14 QUESTIONS

REMEMBER The smaller the percentage error, the more accurate the measurement. Use the instrument's precision when it is given.

ROUND 1 Questions 1–8: find the percentage error, correct to 2 decimal places.

1. 28 m
2. 3.6 cm
3. 12.45 kg
4. 175 cm
5. 0.7 L
6. 2.35 m
7. 48 s
8. 9.5 kg

ROUND 2 Questions 9–12: find the percentage error, correct to 2 decimal places.

9. 350 g on scales marked every 10 g
10. 75 mL in a jug marked every 25 mL
11. 60 km/h on a speedo marked every 10 km/h
12. 1.2 m on a tape marked every 5 cm

ROUND 3 Questions 13–14: solve each problem, correct to 2 decimal places. Show your working.

13. Mia's pencil is 15 cm long and she is 168 cm tall, both to the nearest cm. Which is more accurate?
14. A 100 m sprint is timed as 12.4 s. Find the percentage error of the time.

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

8 QUESTIONS

1. Find the percentage error of 8 m, to the nearest metre. [10.3]
2. Write the limits of 5.3 kg. [10.2]
3. A scale marked every 20 g shows 400 g. Find the percentage error. [10.3]
4. Convert 4 500 g to kg. [10.1]
5. Find the percentage error of 0.5 L. [10.3]
6. Which is more accurate: 10 m or 100 m, each to the nearest metre? [10.3]
7. Find the area of a triangle with base 10 cm and height 6 cm. [Year 8]
8. Find the percentage error of 1.25 cm. [10.3]

LESSON 10.4

Perimeter and area of composite shapes

Date _____

Learning intention
We are learning to find the perimeter and area of shapes made from rectangles and triangles.

Success criteria
- ☐ I can find missing side lengths, then the perimeter
- ☐ I can find an area by splitting a shape into rectangles and triangles
- ☐ I can find a shaded area by subtracting

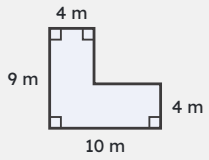
Level 1 · Getting started

Easy · 12 questions

WE DO With your teacher: copy what they write on the board.

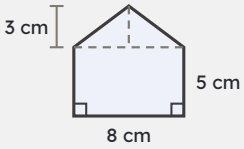
Example 1 Find the perimeter or area. Find any missing sides first.

a Perimeter

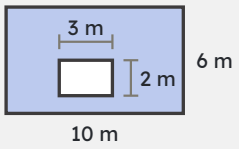


b Area of the shape in part a

c Area



d Shaded area



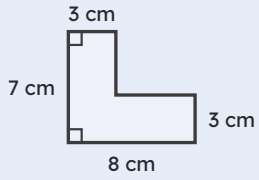
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

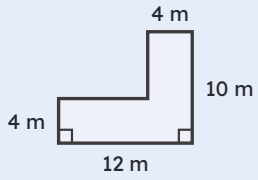
REMEMBER Rectangle A = l × w · Triangle A = ½ × b × h · Shaded area = whole shape − the white part · Perimeter: add every side.

ROUND 1 Questions 1–4: find the perimeter of each shape.

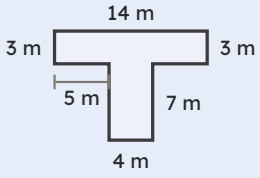
1



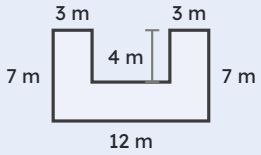
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3

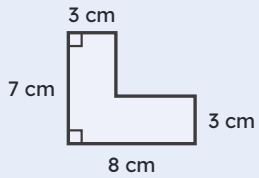


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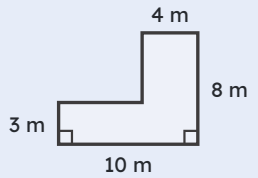


ROUND 2 Questions 5–8: find the area of each shape.

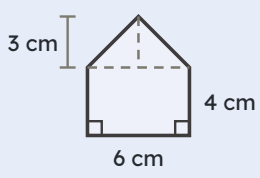
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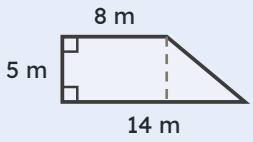
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7

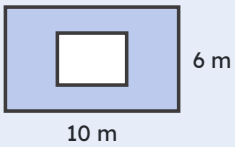


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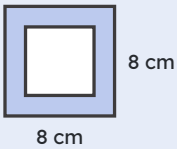


ROUND 3 Questions 9–12: find each shaded area.

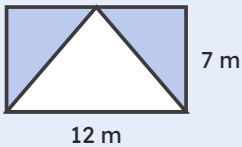
9 White: 4 m by 3 m



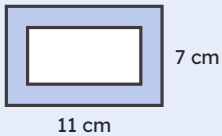
10 White square: 5 cm



11



12 White: 8 cm by 4 cm



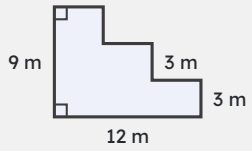
Level 2 · Building up

Medium · 8 questions

WE DO With your teacher: copy what they write on the board.

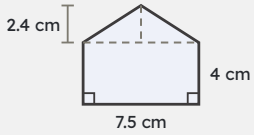
Example 2 **Find the perimeter or area. Split the shape, or subtract.**

a Perimeter

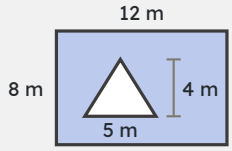


b Area of the shape in part a. Each step is 4 m wide.

c Area



d Shaded area



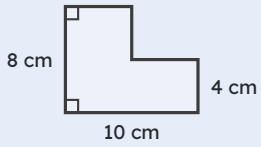
YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line.

8 QUESTIONS

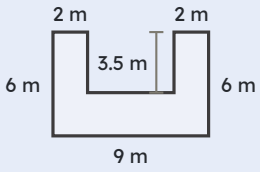
REMEMBER Missing sides: use the opposite sides (across the shape the lengths add up). Split into rectangles and triangles, or take away.

ROUND 1 Questions 1–4: find the perimeter of each shape.

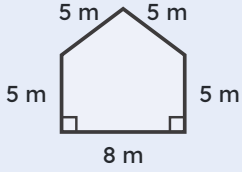
1



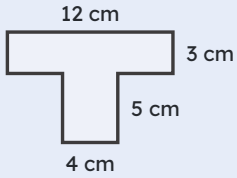
2



3

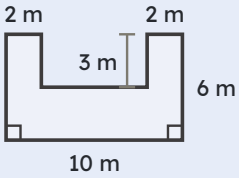


4 It is symmetrical.

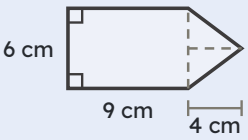


ROUND 2 Questions 5–8: find the area of each shape.

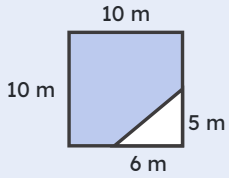
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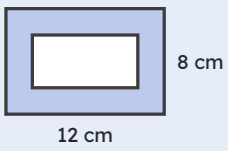
6



7



8 White: 8 cm by 4 cm



MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

8 QUESTIONS

1. A rectangle is 12 m by 5 m. Find its perimeter. (10.4)
2. Find the percentage error of 40 cm, to the nearest cm. (10.3)
3. A 12 m by 8 m yard has a 3 m by 2 m garden bed; the rest is lawn. Lawn area? (10.4)
4. Convert 2.4 m to cm. (10.1)
5. A triangle has base 9 m and height 4 m. Find its area. (10.4)
6. Write the limits of 7.2 m. (10.2)
7. An L-shape is made of a 6 m by 2 m and a 3 m by 2 m rectangle. Area? (10.4)
8. A square has an area of 49 cm². Find its perimeter. (10.4)

LESSON 10.5

Areas of quadrilaterals

Date _____

Learning intention
We are learning to find the area of a parallelogram, a trapezium, a rhombus and a kite.

Success criteria
- ☐ I can use A = bh for a parallelogram
- ☐ I can use A = $\frac{1}{2}$(a + b)h for a trapezium
- ☐ I can use A = $\frac{1}{2}$xy for a rhombus or a kite

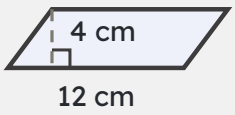
Level 1 · Getting started

Easy · 12 questions

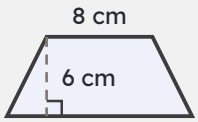
WE DO With your teacher: copy what they write on the board.

Example 1 Find the area of each shape. Write the formula first.

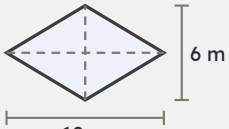
a Parallelogram



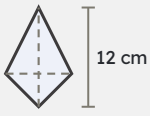
b Trapezium



c Rhombus



d Kite



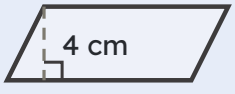
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

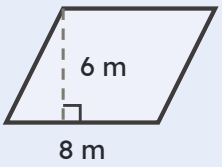
REMEMBER h is the perpendicular height (the dashed line) · a and b are the parallel sides · x and y are the diagonals.

ROUND 1 Questions 1–4: find the area of each parallelogram.

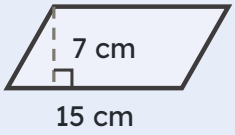
1.



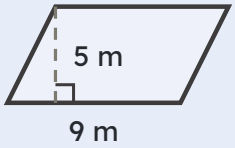
2.



3.

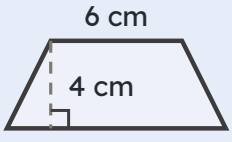


4.

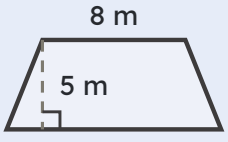


ROUND 2 Questions 5–8: find the area of each trapezium.

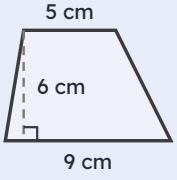
5.



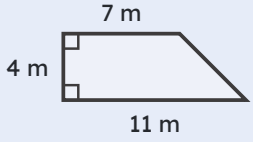
6.



7.

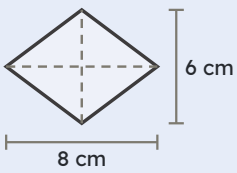


8.

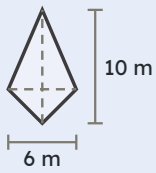


ROUND 3 Questions 9–12: find the area of each rhombus or kite.

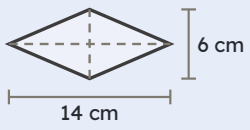
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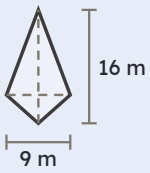
10.



11.



12.



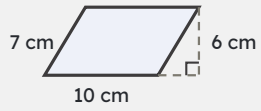
Level 2 · Building up

Medium · 8 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **Choose the right formula. Some measurements are not needed.**

a Area



b A trapezium has parallel sides 7.5 m and 4.5 m, 3.2 m apart. Find its area.

c A parallelogram has an area of 54 cm² and a base of 9 cm. Find its height.

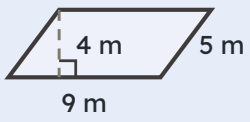
d A kite has diagonals of 1.2 m and 0.8 m. Find its area.

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes). 8 QUESTIONS

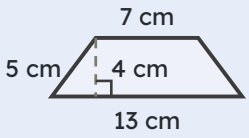
REMEMBER The slanted side is not the height · Missing length: put the numbers into the formula, then work backwards.

ROUND 1 Questions 1–4: find the area of each shape. One length in each is not needed.

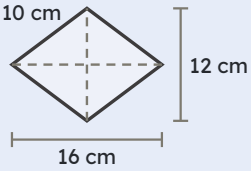
1



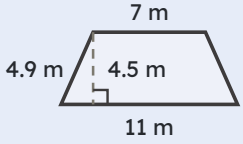
2



3



4



ROUND 2 Questions 5–8: find the missing measurement. Show your working.

5 A parallelogram has an area of 72 cm² and a base of 9 cm. Find its height.

6 A rhombus has an area of 24 m² and one diagonal of 8 m. Find the other diagonal.

7 A trapezium has parallel sides of 6 cm and 10 cm and an area of 40 cm². Find its height.

8 A square has diagonals of 10 cm. Use the rhombus formula to find its area.

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind. 8 QUESTIONS

1 Find the area of a parallelogram with base 7 m and height 3 m. [10.5]

2 Find the perimeter of a rectangle 9 cm by 4 cm. [10.4]

3 A trapezium has parallel sides 4 cm and 6 cm, 5 cm apart. Area? [10.5]

4 Percentage error of 16 kg, to the nearest kg? [10.3]

5 A rhombus has diagonals of 9 m and 4 m. Find its area. [10.5]

6 Convert 0.6 km to m. [10.1]

7 A kite has diagonals of 10 cm and 7 cm. Find its area. [10.5]

8 A parallelogram has an area of 30 m² and a base of 6 m. Height? [10.5]

LESSON 10.6

Circumference and arc length

Date _____

Learning intention
We are learning to find circumferences, arc lengths and perimeters of curved shapes.

Success criteria
- ☐ I can use $C = \pi d$ or $C = 2\pi r$ to find a circumference
- ☐ I can find an arc length: $\frac{\theta}{360} \times 2\pi r$
- ☐ I can find the perimeter of sectors and composite shapes

Level 1 · Getting started

Easy · 20 questions

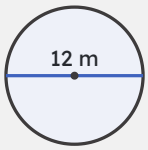
WE DO With your teacher: copy what they write on the board.

Example 1 Find each circumference or perimeter, correct to one decimal place.

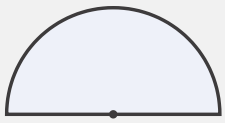
a Circumference



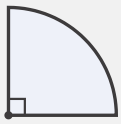
b Circumference



c Perimeter



d Perimeter



YOU DO On your own: Easy work down the box, one round at a time. Write each answer on its line.

20 QUESTIONS

REMEMBER $C = \pi d = 2\pi r$ · Semicircle $P = \frac{1}{2} \times \pi d + d$ · Quadrant $P = \frac{1}{4} \times 2\pi r + r + r$ · Use the π key.

ROUND 1 Questions 1–8: find the circumference of each circle with radius r, correct to one decimal place.

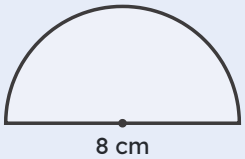
1. r = 3 cm
2. r = 7 m
3. r = 10 cm
4. r = 2.5 m
5. r = 8 mm
6. r = 1.5 km
7. r = 20 cm
8. r = 6.5 m

ROUND 2 Questions 9–16: find the circumference of each circle with diameter d, correct to one decimal place.

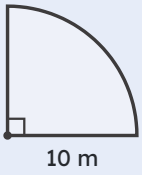
9. d = 8 cm
10. d = 15 m
11. d = 30 mm
12. d = 2 m
13. d = 9 cm
14. d = 50 cm
15. d = 1.4 km
16. d = 25 m

ROUND 3 Questions 17–20: find the perimeter of each shape, correct to one decimal place.

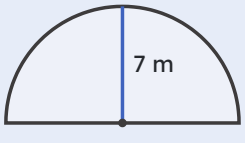
17.



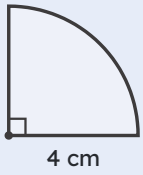
18.



19.



20.



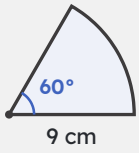
Level 2 · Building up

Medium · 8 questions

WE DO With your teacher: copy what they write on the board.

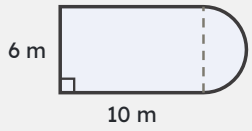
Example 2 **Give each answer correct to one decimal place.**

a Arc length



b Perimeter of the sector in part a

c Perimeter



d Radius of a circle with circumference 50 cm

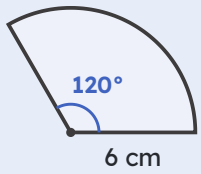
YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line.

8 QUESTIONS

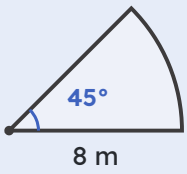
REMEMBER Arc length = $\frac{\theta}{360} \times 2\pi r$ (θ = the angle at the centre) · Sector P = arc + r + r · Composite: add the outside edges, not the dashed lines.

ROUND 1 Questions 1–4: find the perimeter of each sector, correct to one decimal place.

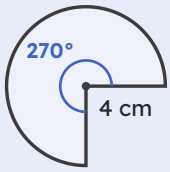
1



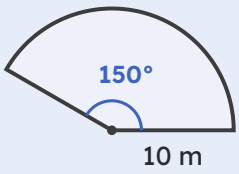
2



3



4

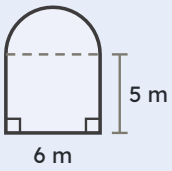


ROUND 2 Questions 5–8: find the perimeter of each shape, correct to one decimal place.

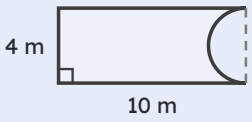
5



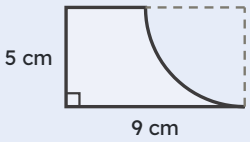
6



7 Cut-out: d = 4 m



8 Cut-out: r = 5 cm



MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

8 QUESTIONS

1 Find the circumference when r = 9 cm. [10.6]

2 Perimeter of a rectangle 7.5 m by 4 m? [10.4]

3 A wheel of diameter 64 cm turns 100 times. How far, in m? [10.6]

4 Convert 345 cm to m. [10.1]

5 A round garden has a circumference of 18.85 m. Radius? [10.6]

6 A kite has diagonals of 6 m and 5 m. Area? [10.5]

7 Find the arc length: 120° sector, r = 3 m. [10.6]

8 Perimeter of a semicircle with d = 20 cm? [10.6]

LESSON 10.7

Areas of circles and sectors

Date _____

Learning intention
We are learning to find the area of circles, semicircles, quadrants, sectors and composite shapes.

Success criteria
- ☐ I can use $A = \pi r^2$ to find the area of a circle
- ☐ I can find the area of a sector: $A = \frac{\theta}{360} \times \pi r^2$
- ☐ I can find composite and shaded areas

Level 1 · Getting started

Easy · 20 questions

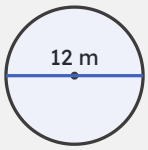
WE DO With your teacher: copy what they write on the board.

Example 1 Find the area of each shape, correct to one decimal place.

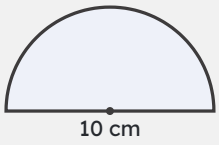
a Circle



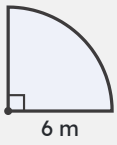
b Circle



c Semicircle



d Quadrant



YOU DO On your own: Easy work down the box, one round at a time. Write each answer on its line.
20 QUESTIONS

REMEMBER $A = \pi r^2$ · $r = d \div 2$ · Semicircle: ½ of a circle · Quadrant: ¼ of a circle · Use the π key, then round.

ROUND 1 Questions 1–8: find the area of each circle, correct to one decimal place.

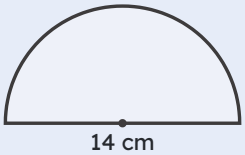
1. r = 3 cm
2. r = 7 m
3. r = 10 mm
4. r = 4 cm
5. r = 2.5 m
6. r = 9 cm
7. r = 1.5 km
8. r = 12 m

ROUND 2 Questions 9–16: find the area of each circle. Halve the diameter first. Correct to one decimal place.

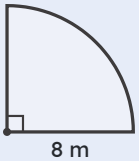
9. d = 16 m
10. d = 22 mm
11. d = 2 m
12. d = 30 cm
13. d = 7 cm
14. d = 26 m
15. d = 50 cm
16. d = 1.2 km

ROUND 3 Questions 17–20: find the area of each shape, correct to one decimal place.

17. Semicircle



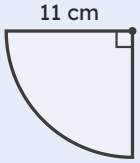
18. Quadrant



19. Semicircle



20. Quadrant



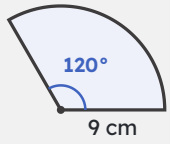
Level 2 · Building up

Medium · 8 questions

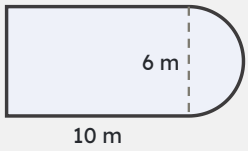
WE DO With your teacher: copy what they write on the board.

Example 2 **Find each area, correct to one decimal place.**

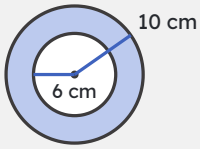
a Sector



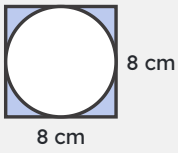
b Area



c Shaded area



d Shaded area



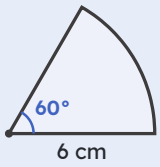
YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line.

8 QUESTIONS

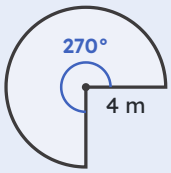
REMEMBER Sector: $A = \frac{\theta}{360} \times \pi r^2$ (θ = the angle at the centre) · Composite: add the parts · Shaded: whole − white part · Ring: $\pi R^2 - \pi r^2$ (R = outer radius, r = inner radius)

ROUND 1 Questions 1–4: find the area of each sector, correct to one decimal place.

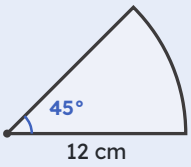
1



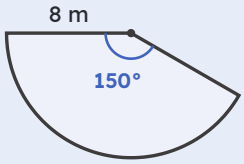
2



3

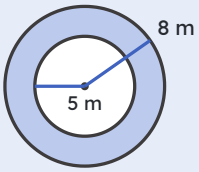


4

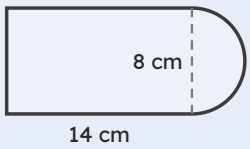


ROUND 2 Questions 5–8: find each area, correct to one decimal place.

5 Shaded area



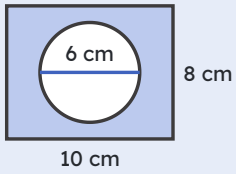
6



7



8 Shaded area



MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

8 QUESTIONS

1. Find the area of a quadrant with radius 4 cm. [10.7]
2. Find the circumference of a circle, d = 10 m. [10.6]
3. A sprinkler waters a circle of radius 4.5 m. Area watered? [10.7]
4. A rhombus has diagonals 8 m and 5 m. Area? [10.5]
5. A pizza of diameter 30 cm is cut into 8 equal slices. Area of one slice? [10.7]
6. A radius is 4 cm, to the nearest cm. Percentage error? [10.3]
7. A circle has area 153.9 cm². Find its radius (÷ π, then √). [10.7]
8. A 10 m by 10 m yard has a round pond of radius 2 m. The rest is lawn. Lawn area? [10.7]

LESSON 10.8

Surface area of a prism

Date _____

Learning intention
We are learning to find the surface area of a prism by adding the areas of its faces.

Success criteria
- ☐ I can find the surface area of a cube and a rectangular prism
- ☐ I can find the surface area of a triangular prism
- ☐ I can find the surface area of an open prism

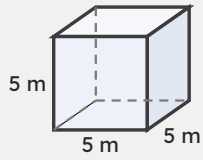
Level 1 · Getting started

Easy · 16 questions

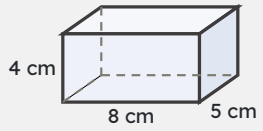
WE DO With your teacher: copy what they write on the board.

Example 1 **Surface area (SA) = the total area of all the faces of a solid.**

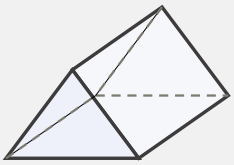
a Cube: find the SA



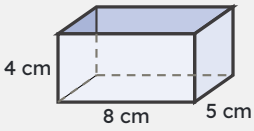
b Find the SA



c How many faces?



d No lid: find the SA



YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

16 QUESTIONS

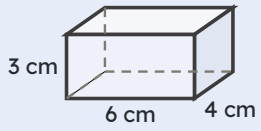
REMEMBER Cube: SA = 6 × side² · Rectangular prism: SA = 2 × (lw + lh + wh) · No lid: take away the top, l × w

ROUND 1 Questions 1–8: find the surface area of each cube with side s.

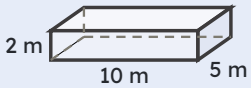
1. s = 2 cm
2. s = 4 m
3. s = 10 mm
4. s = 7 cm
5. s = 1.5 m
6. s = 9 cm
7. s = 3 m
8. s = 20 mm

ROUND 2 Questions 9–12: find the surface area of each rectangular prism.

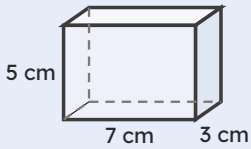
9.



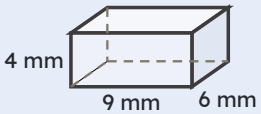
10.



11.

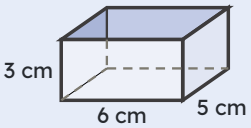


12.

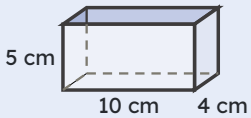


ROUND 3 Questions 13–16: each box has no lid. Find its surface area.

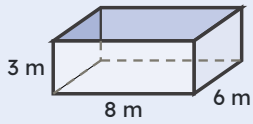
13.



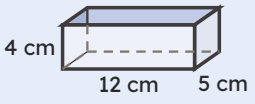
14.



15.



16.



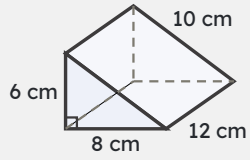
Level 2 · Building up

Medium · 6 questions

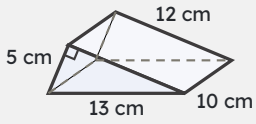
WE DO With your teacher: copy what they write on the board.

Example 2 **Find each surface area. In d, paint a room's 4 walls and ceiling.**

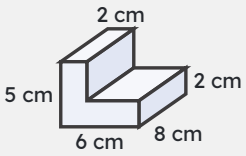
a Triangular prism



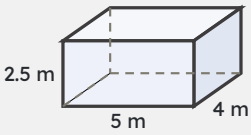
b Triangular prism



c L-shaped prism



d Walls and ceiling

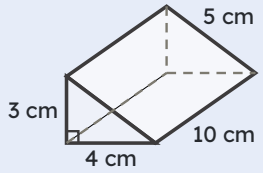


YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes). 6 QUESTIONS

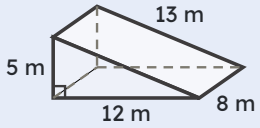
REMEMBER Any prism: SA = 2 × area of the end + perimeter of the end × length · Open: leave out the missing faces.

ROUND 1 Questions 1–4: find the surface area of each triangular prism.

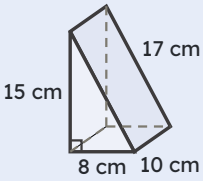
1



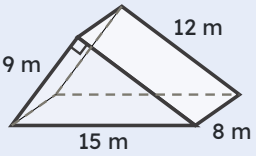
2



3



4



ROUND 2 Questions 5–6: solve each problem. Show your working.

5 A gift box is 30 cm long, 20 cm wide and 10 cm high. How much paper covers it exactly?

6 A fish tank has no lid. It is 60 cm long, 30 cm wide and 40 cm high. What area of glass is used?

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind. 8 QUESTIONS

1 Find the SA of a cube with sides of 8 cm. 10.8

2 Right-angled triangle: 12, 16, 20 cm. Area? 10.4

3 Find the SA of a box 10 cm × 6 cm × 4 cm. 10.8

4 Trapezium: a = 5 m, b = 9 m, h = 4 m. Area? 10.5

5 A cube's SA is 54 m². Side? (÷ 6, then √) 10.8

6 Convert 1.8 m to cm. 10.1

7 A prism is 10 cm long. Its end has area 18 cm² and perimeter 22 cm. Find its SA. 10.8

8 A tank 3 m × 2 m × 1 m high has no lid. SA? 10.8

LESSON 10.9

Surface area of a cylinder

Date _____

Learning intention
We are learning to find a cylinder's surface area: two circles and a rolled-up rectangle.

Success criteria
- ☐ I can find the area of the circular ends: $A = \pi r^2$
- ☐ I can find the curved surface area: $A = 2\pi rh$
- ☐ I can find the surface area of closed and open cylinders

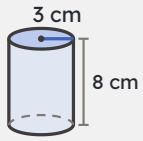
Level 1 · Getting started

Easy · 16 questions

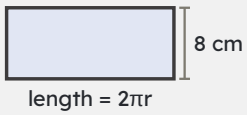
WE DO With your teacher: copy what they write on the board.

Example 1 **Find each area, correct to one decimal place.**

a Area of the top

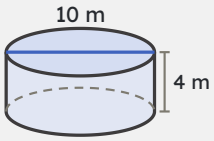


b Curved surface of part a, unrolled



c Total surface area of the cylinder in part a

d Total surface area



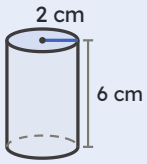
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.
16 QUESTIONS

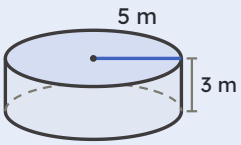
REMEMBER Each end: πr^2 · Curved surface: $2\pi rh$ (unrolled, a rectangle $2\pi r$ by h) · Closed cylinder: $SA = 2\pi r^2 + 2\pi rh$

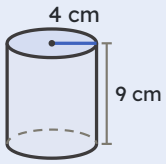
ROUND 1 Questions 1–8: find the curved surface area ($2\pi rh$), correct to one decimal place.

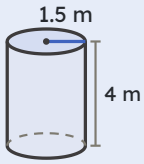
1. $r = 2$ cm, $h = 5$ cm
2. $r = 5$ m, $h = 6$ m
3. $r = 1$ m, $h = 9$ m
4. $r = 7$ mm, $h = 3$ mm
5. $r = 2.5$ cm, $h = 8$ cm
6. $r = 10$ cm, $h = 12$ cm
7. $r = 1.5$ m, $h = 2$ m
8. $r = 6$ m, $h = 2.5$ m

ROUND 2 Questions 9–12: find the total surface area of each closed cylinder, correct to one decimal place.

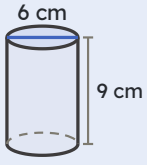
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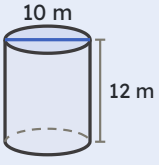
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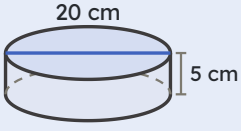
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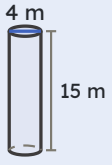
12.


ROUND 3 Questions 13–16: find the total surface area (all closed, $r = d \div 2$), correct to one decimal place.

13.


14.


15.


16.


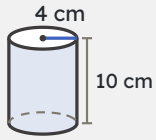
Level 2 · Building up

Medium · 6 questions

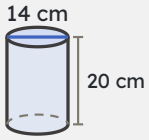
WE DO With your teacher: copy what they write on the board.

Example 2 **Find each area, correct to one decimal place. List the surfaces first.**

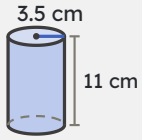
a No lid



b Closed



c The label



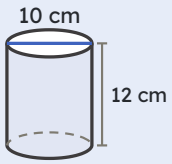
d A pipe open at both ends has r = 2 cm and length 50 cm. Find its outside area.

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes). 6 QUESTIONS

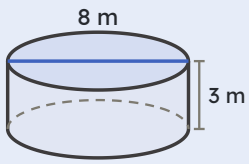
REMEMBER No lid: $\pi r^2 + 2\pi rh$ · Open at both ends, or a label: $2\pi rh$ only · Closed: $2\pi r^2 + 2\pi rh$ · r = d ÷ 2

ROUND 1 **Questions 1–4: find the total surface area, correct to one decimal place. Some are open.**

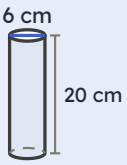
1 No lid



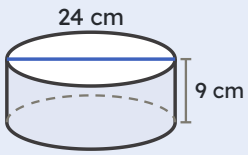
2 Closed



3 Open both ends



4 No lid



ROUND 2 **Questions 5–6: solve each problem. Show your working.**

5 A closed tank is a cylinder, d = 2 m, h = 1.5 m. Paint costs $12 per m². Cost to paint it?

6 A paint roller, d = 6 cm, is 25 cm long. Find the area it paints in 10 turns, to the nearest cm².

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind. 8 QUESTIONS

1 Curved surface area if d = 10 cm, h = 7 cm? 10.9

2 Find the area of a circle with diameter 13 cm. 10.7

3 Closed tin, r = 6 cm, h = 10 cm. Surface area? 10.9

4 Surface area of a 5 cm by 4 cm by 2 cm box? 10.8

5 Bucket (no lid), d = 30 cm, h = 25 cm. SA? 10.9

6 A can is 11.5 cm tall. Write the limits. 10.2

7 Curved SA = 314.2 cm², h = 10 cm. r? (÷ 2πh) 10.9

8 Which has more curved surface: r = 2 cm, h = 6 cm or r = 3 cm, h = 4 cm? 10.9

LESSON 10.10

Volumes of prisms and cylinders

Date _____

Learning intention
We are learning to find the volume of a prism or a cylinder (V = Ah), and its capacity.

Success criteria
- ☐ I can use V = Ah to find the volume of any prism
- ☐ I can use V = πr²h to find the volume of a cylinder
- ☐ I can change between cm³, mL, L and m³

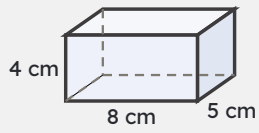
Level 1 · Getting started

Easy · 16 questions

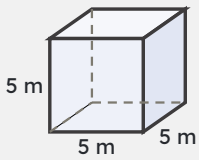
WE DO With your teacher: copy what they write on the board.

Example 1 **Find each volume: V = Ah. The cylinder: correct to one decimal place.**

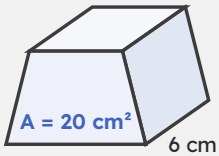
a Rectangular prism



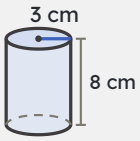
b Cube



c A = area of the end



d Cylinder



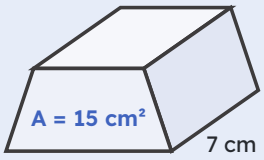
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.
16 QUESTIONS

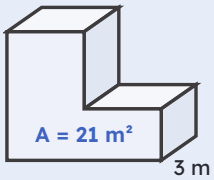
REMEMBER Any prism: V = Ah (A = area of the end, h = length between the ends) · Box: V = l × w × h · Cube: V = s³ · Cylinder: V = πr²h

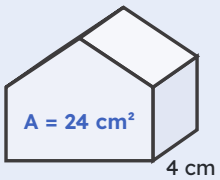
ROUND 1 Questions 1–8: find the volume of each rectangular prism or cube.

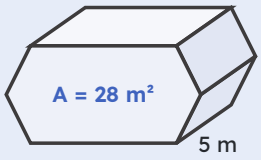
1. 6 cm × 4 cm × 3 cm
2. 10 m × 5 m × 2 m
3. cube, side 3 cm
4. 8 mm × 5 mm × 5 mm
5. 12 cm × 10 cm × 4 cm
6. cube, side 6 m
7. 2.5 m × 4 m × 3 m
8. cube, side 10 cm

ROUND 2 Questions 9–12: A is the area of the end. Find the volume of each prism.

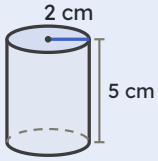
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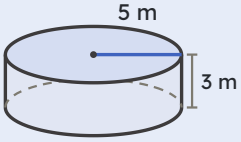
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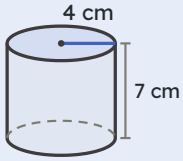
11. 

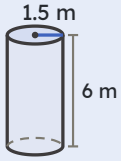
12. 

ROUND 3 Questions 13–16: find the volume of each cylinder, correct to one decimal place.

13. 

14. 

15. 

16. 

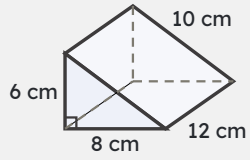
Level 2 · Building up

Medium · 6 questions

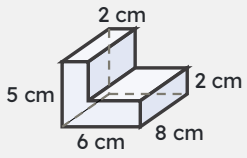
WE DO With your teacher: copy what they write on the board.

Example 2 **Find the area of the end first. 1 cm³ = 1 mL and 1000 cm³ = 1 L.**

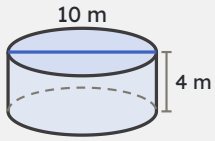
a Triangular prism



b L-shaped prism



c Cylinder (one decimal place)



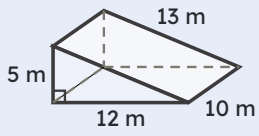
d A fish tank is 50 cm long, 30 cm wide and 40 cm high. Find its capacity in litres.

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes). 6 QUESTIONS

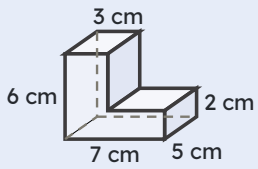
REMEMBER Triangle end: A = ½ × base × height · L-shape: split into rectangles · r = d ÷ 2 · 1 cm³ = 1 mL · 1 m³ = 1000 L

ROUND 1 Questions 1–4: find the volume of each prism or cylinder. Cylinders: correct to one decimal place.

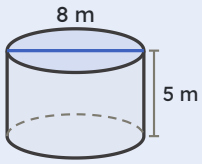
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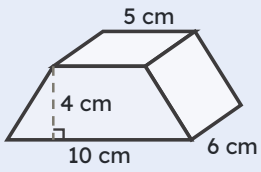
2



3



4



ROUND 2 Questions 5–6: solve each problem. Show your working.

5 A cylindrical water tank has diameter 3 m and height 2 m. Find its capacity, to the nearest litre.

6 A pool is 25 m long, 10 m wide and 1.5 m deep. How many litres of water does it hold?

MIXED PRACTICE Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind. 8 QUESTIONS

1 Volume of a 10 cm × 6 cm × 5 cm box? 10.10

2 Find the area of a circle with radius 5.5 cm. 10.7

3 Prism: end area 16 m², length 5 m. Volume? 10.10

4 Closed tank, r = 2 m, h = 5 m. Surface area? 10.9

5 A can: d = 6 cm, h = 10 cm. Capacity in mL? 10.10

6 Convert 3.5 L to mL. 10.1

7 A box: 120 cm³, base 6 cm × 4 cm. Height? 10.10

8 Carton 8 cm × 5 cm × 25 cm. Capacity in L? 10.10